

YUCHEN LI

Class of 2027 · Target Role: Java Backend Engineering Intern / Agent Engineering Intern
Available to start soon · 5 days/week · 6 months · Email: edllyc041202@gmail.com

EDUCATION

Anhui University – B.Eng. in Digital Media Technology

Sep 2023 – Jun 2027

- **Research & Awards:** First-author student paper under external review at *IEEE Transactions on Medical Imaging (IEEE TMI)*, proposing a topology-prior connectivity loss function; awarded **Second Prize** in the Mathematical Contest in Modeling, **Third Prize** in the Provincial Mathematical Modeling Competition, and a **Third-Class Academic Scholarship**.
- **English Proficiency:** **IELTS 7.0**; able to read and reproduce English technical documentation and academic papers.
- **Academic Performance:** Average score **87.6/100**; completed core CS coursework including Data Structures, Operating Systems, Computer Networks, and Databases.

TECHNICAL SKILLS

- **Java / JVM / Concurrency:** Solid foundation in Java, including collections, concurrent programming, and the JVM memory model; experienced in asynchronous orchestration and high-concurrency workflow design using **CompletableFuture**.
- **Spring Boot / MyBatis-Plus / MySQL:** Proficient in backend service development with **Spring Boot**, **MyBatis-Plus**, and **MySQL**; hands-on experience with index optimization, transaction isolation, SQL tuning, and persistence-layer implementation in production-style business flows.
- **Redis / Redisson / Lua / RocketMQ / Sentinel / Docker / Maven / Git:** Experienced with Redis-based caching and concurrency control using **Redis**, **Redisson**, and **Lua**; familiar with multi-level caching, distributed locking, atomic inventory deduction, MQ transactional and delayed messaging, as well as rate limiting and circuit-breaking; comfortable with basic **Docker** containerization and day-to-day engineering collaboration using **Maven** and **Git**.
- **LangChain4j / WebFlux / Reactor / RAG / Ollama / RAGAS / LLM-as-a-Judge:** Experienced in building agent applications with **LangChain4j**, **Spring WebFlux**, and **Project Reactor**; familiar with AiServices-driven entity extraction, query rewriting, intent routing, and complexity assessment; hands-on with **Plan-and-Execute**, **Parent-Child** chunking, hybrid retrieval with **Milvus/OpenSearch**, **RRF**, **BM25+**, reranking, and local/offline evaluation using **Ollama**, **RAGAS**, and **LLM-as-a-Judge**.

PROJECT EXPERIENCE

12306 High-Concurrency Ticketing Core Gateway (Backend Architecture Development)

Jan 2026 – May 2026

Tech Stack: Spring Boot, Redis, Redisson, Caffeine, RBloomFilter, RocketMQ, Sentinel, MyBatis-Plus

Project Summary: Built core workflows for a 12306-style high-concurrency ticketing system, covering ticket query, booking, and refund scenarios, with a focus on cache governance, inventory pre-deduction, transactional consistency, and closed-loop compensation.

Key Contributions:

1. **Caching and High-Concurrency Governance:** Built a multi-level cache with **Caffeine + Redis + RBloomFilter**; combined **Redisson** distributed locks with **CompletableFuture**-based singleflight request coalescing to handle hot-key rebuilds. In load testing on the ticket-query hot path, the system reached **5409.54 QPS** at **800 concurrent users**, with **0% HTTP error rate** and **344 ms P95 latency**.
2. **Inventory Deduction and Segmented Stock Model:** Implemented atomic inventory pre-deduction using **Redis Lua**; designed a **Segmented Stock** model to spread hotspot traffic and reduce inventory contention and overselling risk under high-concurrency booking scenarios.
3. **Asynchronous Ordering and Transactional Messaging:** Connected Redis pre-deduction, order persistence, and MySQL inventory deduction through **RocketMQ transactional messages** and **TransactionListener**, forming an asynchronous peak-shaving and eventual-consistency pipeline. In a booking load test with **200 users**, the system achieved **510.94 TPS**, **0% error rate**, and **318 ms P95 latency**; no overselling was observed, and post-async consistency checks passed.
4. **Refund Compensation and Stability Protection:** Designed a refund workflow using a **refund** status field and an idempotent **Redis Lua** compensation mechanism to prevent duplicate stock restoration caused by MQ retries; implemented automatic closure of unpaid timed-out orders with **RocketMQ delayed messages**; added **Sentinel**-based rate limiting and hotspot protection.

12306 Intelligent Digital Employee (Reactive Multi-Agent / RAG System Development)

Jan 2026 – May 2026

Tech Stack: Spring WebFlux, Project Reactor, LangChain4j, Milvus, OpenSearch, Zhipu GLM, Ollama, RAGAS

Project Summary: Built a reactive multi-agent and RAG execution framework for 12306 policy Q&A and ticket booking/refund/change/cancellation scenarios, covering intent routing, plan execution, hybrid retrieval, memory management, and streaming response delivery.

Key Contributions:

1. **Multi-Expert Intent Routing:** Designed a multi-expert intent scoring mechanism to score **ACTION**, **TICKET**, **RAG**, and **CHITCHAT** intents in parallel; combined top score, score margin, and confidence thresholds to make layered routing decisions, reducing misrouting across action requests, ticket queries, and policy Q&A.
2. **Policy Routing and Semantic Bridging:** Built context-dependency detection, policy classification, and policy-evidence judgment pipelines with **AiServices**; combined **session-prefix isolation**, **ConversationBridge**, and locally deployed lightweight models via **Ollama** to improve follow-up continuity and semantic consistency across expert switching.

3. **Plan-and-Execute Workflow:** Upgraded the traditional ReAct pattern to **Plan-and-Execute**; used a Planner to output JSON plans for decomposing complex booking and refund requests, while an Executor invoked ticketing tools and consumed structured **ToolObservation** results; added a **Replan** mechanism to recover from execution-time exceptions.
4. **Hybrid Retrieval and Planned RAG:** Applied **Parent-Child** chunking together with dual-channel retrieval using **Milvus Dense Retrieval + OpenSearch BM25**; improved answer quality for complex policy Q&A with **RRF** fusion, **BM25+** scoring, and reranking; introduced sub-question decomposition for complex queries to support **Planned RAG**.
5. **Evaluation, Streaming Interaction, and Memory:** Ran **RAGAS** evaluation on a curated benchmark set, achieving **0.881 context precision** and **0.935 context recall**; implemented streaming output with **WebFlux + Project Reactor + Sinks.Many**; integrated Redis-based semantic caching and a dual-layer memory design combining a **short-term sliding window** with **long-term summary compression**.